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DSC624 / ASSIGNMENT 1 / TASK 2

Hybrid Supply-Chain Late-Delivery Risk Prediction

A controlled comparison of numerical features, TF-IDF, and transformer embeddings

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Abstract

This study evaluates whether pretrained sentence-transformer embeddings improve early prediction of late-delivery risk in the DataCo SMART Supply Chain dataset [1]. The experiment compares three identically configured Random Forest classifiers: numerical data only; numerical data plus TF-IDF features; and numerical data plus all-MiniLM-L6-v2 sentence embeddings [2, 3]. An order-disjoint chronological holdout contains 9,600 training and 2,400 test records. The transformer hybrid achieves F1 = 0.6839 against 0.6742 for the numerical baseline, with improved recall (0.6217 versus 0.5840), but lower accuracy and precision. A paired bootstrap estimates an F1 difference of +0.0097 (95% interval: -0.0042 to +0.0221). The evidence therefore suggests a modest recall benefit, not a conclusive overall performance improvement. The contribution is an ablation-controlled assessment of semantic features, including the possibility that the embedding model is redundant.

1. Introduction and research question

Late shipments create avoidable operational and customer-service costs. A useful risk detector should flag orders likely to arrive late using information plausibly available when an order is placed, rather than learning from observed delivery outcomes. This report addresses one question: Does adding pretrained transformer representations of order and shipment metadata improve late-delivery-risk classification over a numerical-only model? A TF-IDF hybrid serves as a non-transformer text control. Although the course task includes fraud detection, this dataset experiment evaluates delivery risk only; no fraud labels are used.

2. Dataset and experimental design

The sole dataset is DataCo SMART Supply Chain for Big Data Analysis [1] (180,519 published records, 53 attributes). Following target/date validity checks, the notebook separates orders using the earliest date associated with each Order Id, taking approximately the first 80% of orders for training and the remaining 20% for testing. Random subsampling within these already separated partitions caps the executed experiment at 12,000 records. The chronological cutoff is 09 July 2017, 23:09:48.

Partition	Records	Unique orders	Late-risk prevalence
Train	9,600	8,852	55.28%
Test	2,400	2,184	53.08%

Table 1. Chronological, order-disjoint sample used for the reported experiment (random seed 42).

The label is the binary Late_delivery_risk field. Numerical predictors (nine) capture scheduled shipping days, item quantity, prices, discounts, item total, latitude and longitude. Text inputs (ten) concatenate product/category/department names, shipping mode, customer segment, destination city/state/country/region, and market into a labeled metadata string. These fields are mainly structured categorical descriptions, not narrative incident reports.

Leakage control: the model excludes the target itself, Delivery Status, actual shipping days, shipping timestamp, Order Status, Benefit per order, and Order Profit Per Order. Split membership is determined before subsampling,

and every Order Id belongs to only one partition. Median imputation and text dimensionality-reduction mappings are fitted exclusively on training records.

3. Modeling methodology

A — Numerical baseline. A Random Forest is trained on nine numerical predictors after median imputation.

B — TF-IDF control. The same numerical features are appended to unigram/bigram TF-IDF representations (maximum 3,000 terms), reduced to 32 dimensions using training-fitted TruncatedSVD.

C — Transformer enhancement. The ten metadata fields are converted into a labeled text sequence. The pretrained all-MiniLM-L6-v2 sentence transformer encodes it as a normalized 384-dimensional vector [3]; training-fitted PCA reduces this to 32 dimensions, which are appended to the identical numerical features.

All three Random Forests use 120 trees, maximum depth 16, minimum leaf size 2, balanced-subsample class weighting, and seed 42. The embedding model is not fine-tuned; it is a pretrained transformer encoder rather than a generative chat LLM. This directly implements the pre-trained embedding extension specified for Task 2.

The same fixed 0.5 decision threshold and held-out test records are used throughout. F1 is the primary selection metric; precision, recall, accuracy, ROC-AUC, and average precision (AP) provide complementary views. A 500-replicate paired row-level bootstrap estimates uncertainty in the transformer-versus-baseline F1 difference.

4. Quantitative results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	AP
Numeric only	0.7004	0.7974	0.5840	0.6742	0.7326	0.7882
Numeric + TF-IDF	0.6850	0.7580	0.5973	0.6681	0.7326	0.7845
Numeric + transformer	**0.6950**	**0.7601**	**0.6217**	**0.6839**	**0.7306**	**0.7747**

Table 2. Held-out results on the same 2,400 test records. F1 is the primary metric; AP = average precision. Highlighted row is the transformer hybrid, not an overall winner on every metric.

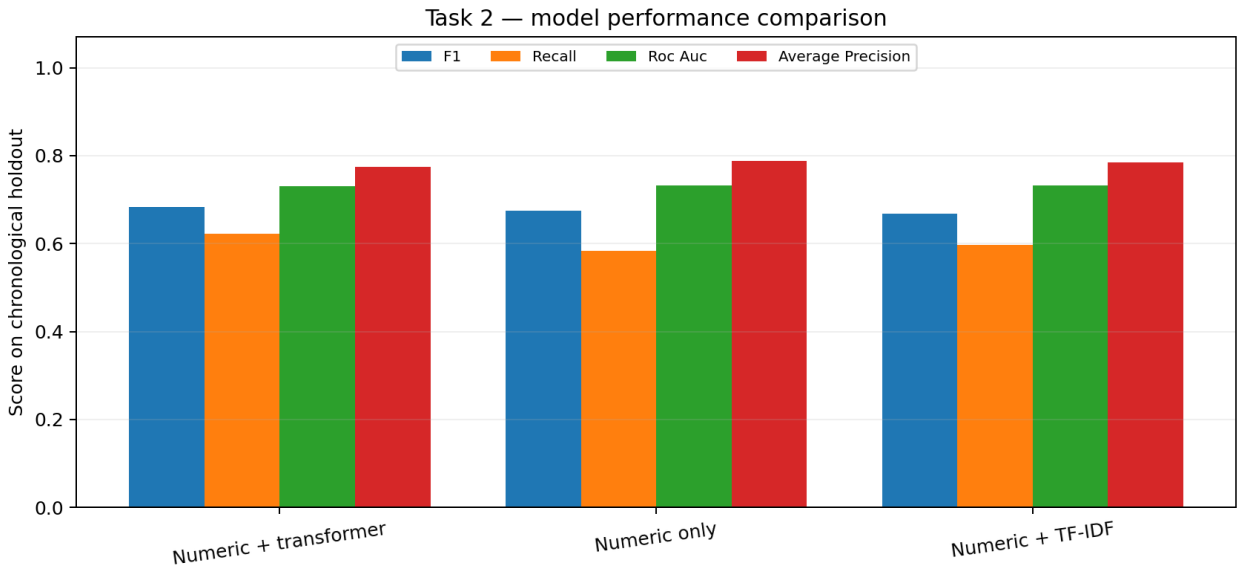


Figure 1. Comparison across four evaluation metrics (source: the executed Colab experiment).

5. Error analysis and performance trade-offs

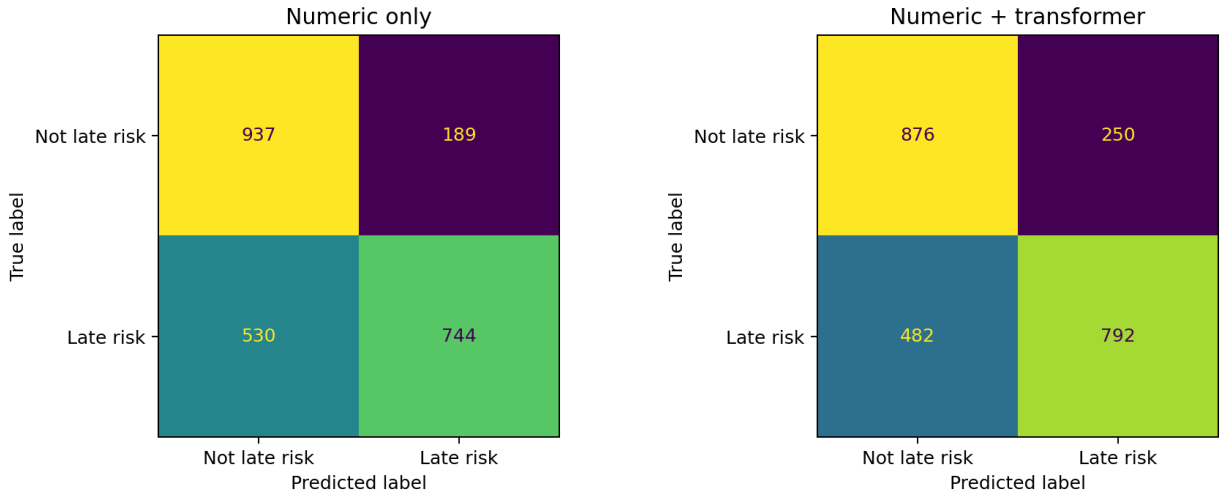


Figure 2. Test-set confusion matrices for the numerical baseline (left) and transformer hybrid (right); the positive class is late-delivery risk.

The numerical baseline identifies 744 of 1,274 late-risk records (530 missed) and incorrectly flags 189 non-late records. The transformer hybrid identifies 792 late-risk records (482 missed) but incorrectly flags 250 non-late records. Relative to the baseline, it detects 48 more late-risk records while producing 61 additional false alarms. This is a recall–precision trade-off rather than a uniform improvement.

F1 rises from 0.6742 to 0.6839: +0.0097 absolute (+1.44% relative), while recall rises by 3.77 percentage points. Precision decreases by 3.74 points and accuracy by 0.54 points. ROC-AUC falls slightly (0.7326 to 0.7306); AP declines from 0.7882 to 0.7747. The TF-IDF control reaches F1 = 0.6681, below both alternatives. Consequently, semantic embeddings perform better than TF-IDF on this specific F1 measure, but a broad ranking advantage is not demonstrated.

6. Interpretation: novelty and statistical evidence

The methodological contribution is fusion of numerical shipping fields and pretrained semantic vectors while holding the classifier and added feature dimension constant. TF-IDF controls for the simpler benefit of adding textual features. Because DataCo largely contains short category and location labels rather than free-text shipping comments, the measured difference cannot be attributed unequivocally to deep semantic reasoning.

The paired F1 gain (+0.0097) has a 95% bootstrap interval of [-0.0042, +0.0221]. Because the interval crosses zero, a small loss or benefit remains plausible; superiority is not established. The hybrid may nevertheless suit recall-oriented triage when missing risky shipments is more expensive than investigating false alarms.

7. Limitations and threats to validity

Task scope. The target is historical late-delivery risk, not a fraud label.

Representativeness. The 12,000-record Colab sample may not represent the entire dataset; sample-size sensitivity remains untested.

Language signal. Text is mostly categorical metadata; shipping mode overlaps with scheduled delivery days.

Uncertainty and dependence. The bootstrap samples rows rather than whole orders; order-cluster resampling would better capture within-order dependence.

Operational constraints. Historical labels, distribution shift, calibration, and false-alarm costs constrain deployment.

8. Reproducibility and conclusion

The accompanying DSC624_Task2_Google_Colab.ipynb downloads the sole Kaggle dataset [1], performs the order-disjoint split, runs all three classifiers, and exports metrics, confusion matrices and experiment_summary.json. The

reported run used CPU, scikit-learn 1.6.1, NumPy 2.4.6, seed 42 and a 12,000-record cap. The notebook includes dependency setup; its pre-trained encoder is sentence-transformers/all-MiniLM-L6-v2.

Transformer embeddings modestly improve F1 and recall but reduce precision, raising false alarms. As the confidence interval crosses zero, an overall advantage is unproven. Future work should test the full dataset, order-cluster intervals, calibrated cost-aware thresholds, and genuine shipping comments.

References

[1] DataCo SMART Supply Chain for Big Data Analysis. Kaggle. https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

[2] Reimers, N. & Gurevych, I. (2019). Sentence-BERT. EMNLP-IJCNLP, 3982-3992. https://aclanthology.org/D19-1410/

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[4] Breiman, L. (2001). Random Forests. Machine Learning 45, 5-32. https://doi.org/10.1023/A:1010933404324